

Aminoguanidine

Aminoguanidine is the chemical compound with the formula $\text{NH}_2\text{C}(=\text{NH})\text{NHNH}_2$. It is a versatile synthetic intermediate.^[1]

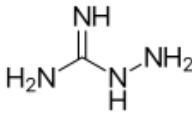
Aminoguanidine was studied as an investigational drug for the treatment of diabetic nephropathy, under the nonproprietary name **Pimagedine**. It is no longer under development as a drug.^[2] Pimagedine functions as an inhibitor of diamine oxidase and nitric oxide synthase. It acts to reduce levels of advanced glycation end products (AGEs) through interacting with 3-deoxyglucosone, glyoxal, methylglyoxal, and related dicarbonyls. These reactive species are converted to less reactive heterocycles by this condensation reaction.

Chemistry

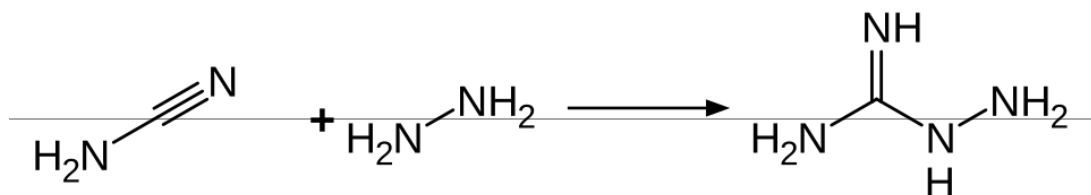
Synthesis

The industrial synthesis uses the reaction between cyanamide and hydrazine in aqueous solution.^[1]

Aminoguanidine

 	
Names	
IUPAC name	
2-Aminoguanidine	
Other names	
Aminoguanidine	
Guanyl hydrazine	
Hydrazinecarboximidamide	
Imino semicarbazide	
Monoaminoguanidine	
Pimagedine	
Identifiers	
CAS Number	79-17-4 (https://commonchemistry.cas.org/detail?cas_rn=79-17-4) ✓
3D model (JSmol)	Interactive image (https://chemapps.stolaf.edu/jmol/jmol.php?model=NC%28N%29N%3DN)
ChEMBL	ChEMBL225304 (https://www.ebi.ac.uk/chembl/explore/compound/ChEMBL225304) ✓
ChemSpider	2061 (https://www.chemspider.com/Chemical-Structure.2061.html) ✗
ECHA InfoCard	100.001.076 (https://echa.europa.eu/substance-information/-/substanceinfo/100.001.076)
IUPHAR/BPS	5135 (http://www.guidetopharmacology.org/GRAC/LigandDisplayForward?tab=summary&ligandId=5135)
KEGG	D05479 (https://www.kegg.jp/entry/D05479)

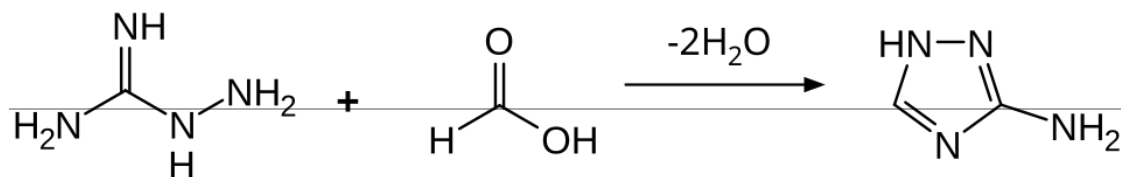
PubChem CID	2146 (https://pubchem.ncbi.nlm.nih.gov/compound/2146)
UNII	SCQ4EZQ113 (https://precision.fda.gov/uniisearch/srs/unii/SCQ4EZQ113) ✗
CompTox Dashboard (EPA)	DTXSID5040964 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID5040964)
InChI	InChI=1S/CH6N4/c2-1(3)5-4/h1,4H,2-3H2 ✗ Key: DRCHNMARIZRTKI-UHFFFAOYSA-N ✗
SMILES	NC(N)=N
Properties	
Chemical formula	CH ₆ N ₄
Molar mass	74.085 g/mol
Density	1.72 g/ml
Boiling point	261 °C (502 °F; 534 K)
log P	-1.475
Related compounds	
Related compounds	Guanidine
Except where otherwise noted, data are given for materials in their <u>standard state</u> (at 25 °C [77 °F], 100 kPa). ✗ verify (what is ✓ ✗ ?) Infobox references	



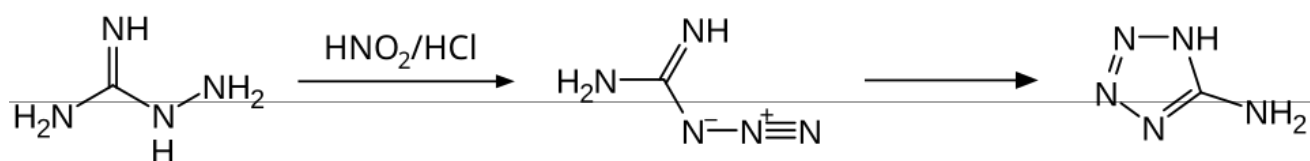
The compound can also be obtained from the reduction of nitroguanidine with zinc in acetic acid.^[3]

Properties

Aminoguanidine is a colorless solid that is soluble in water and ethanol. It is basic, producing salts when reacted with organic acids. As established by many crystallographic studies, protonation of aminoguanidine occurs at the imino nitrogen.^[4] With formic acid, condensation occurs, leading to cyclization to give 3-amino-1,2,4-triazole.^[1]



The compound reacts with nitrous acid in acidic medium to give 5-aminotetrazole via the intermediate guanylazide.^[1] At neutral pH, the reaction leads to tetrazene.^[5] Diazotization in acetic acid yields 1,3-di-(tetrazolyl)-triazene.^[1]



Investigational drug

Pimagedine was under development as a drug for kidney diseases by the pharmaceutical company Alteon (now known Synvista Therapeutics Inc.) that was founded in 1986.^[6] In 1987, Alteon acquired a license to intellectual property relating to AGE inhibition from Rockefeller University.^[7] In 1989, Alteon and Marion Merrell Dow Inc (MMD) entered into a joint development program for pimagedine.^[8] In 1992, Alteon licensed a patent from Rockefeller University relating to the use of pimagedine to inhibit AGE formation.^[7] In 1995, Hoechst AG (now Sanofi-Aventis) acquired MMD and subsequently terminated its agreement with Alteon, which led Alteon to stop clinical trials, which caused some controversy.^[8] In 1997, Alteon and Genentech announced a collaboration under which Genentech would fund development of pimagedine and would have the rights to sell the drug if it would be approved.^[9]

In March 1998, Alteon announced that it had been advised that it should discontinue its Phase III trial of pimagedine in non-insulin-dependent (type II) diabetes patients with overt nephropathy, after the trial's external safety monitoring committee found an increased risk of side effects in the treatment group.^[10] In November 1998, Alteon announced that its Phase III trial for pimagedine as a treatment for end stage renal disease had failed to prove efficacy, which led Carl Gordon, a leading biotech analyst, to say: "It looks like pimagedine is probably finished."^[11] In February, 1999, Genentech ended its collaboration with Alteon to develop pimagedine.^[12] In April 1999 Alteon announced that it would cease development of pimagedine as a treatment for end stage renal disease but might consider continuing development in type 1 diabetic patients with overt nephropathy or progressive kidney disease.^[13] Alteon's 2000, 2001, 2002 annual reports indicated that it was not running any clinical trials on pimagedine but was seeking co-development partners.^{[14][15][16]} Alteon's 2003 and subsequent annual reports did not mention that Alteon was seeking partners for pimagedine,^[17] which indicated that efforts to interest other companies and investors had failed and which signaled that commercial efforts to develop pimagedine as a drug were indeed finished.

References

1. Güthner, Thomas; Mertschenk, Bernd; Schulz, Bernd (2006), "Guanidine and Derivatives", *Ullmann's Encyclopedia of Industrial Chemistry*, American Cancer Society, doi:10.1002/14356007.a12_545.pub2 (https://doi.org/10.1002%2F14356007.a12_545.pub2), ISBN 978-3-527-30673-2
2. Thornalley, Paul J. (2003). "Use of aminoguanidine (Pimagedine) to prevent the formation of advanced glycation endproducts". *Archives of Biochemistry and Biophysics*. **419** (1): 31–40. doi:10.1016/j.abb.2003.08.013 (<https://doi.org/10.1016%2Fj.abb.2003.08.013>). PMID 14568006 (<https://pubmed.ncbi.nlm.nih.gov/14568006>).
3. Smith, G. B. L.; Anzelmi, Edward (1935-12-01). "Reduction of Nitroguanidine. III. Synthesis of Aminoguanidine1". *Journal of the American Chemical Society*. **57** (12): 2730. Bibcode:1935JChS..57.2730S (<https://ui.adsabs.harvard.edu/abs/1935JChS..57.2730S>). doi:10.1021/ja01315a510 (<https://doi.org/10.1021%2Fja01315a510>). ISSN 0002-7863 (<https://search.worldcat.org/issn/0002-7863>).

4. Adams, J. M. (1977). "The Crystal Structure of Aminoguanidinium Dihydrogen Orthophosphate". *Acta Crystallographica Section B Structural Crystallography and Crystal Chemistry*. **33** (5): 1513–1515. Bibcode:1977AcCrB..33.1513A (<https://ui.adsabs.harvard.edu/abs/1977AcCrB..33.1513A>). doi:10.1107/S0567740877006402 (<https://doi.org/10.1107%2FS0567740877006402>).
5. Patinkin, Seymour H.; Horwitz, Jerome P.; Lieber, Eugene (1955-02-01). "The Structure of Tetracene1,2". *Journal of the American Chemical Society*. **77** (3): 562–567. Bibcode:1955JACHS..77..562P (<https://ui.adsabs.harvard.edu/abs/1955JACHS..77..562P>). doi:10.1021/ja01608a014 (<https://doi.org/10.1021%2Fja01608a014>). ISSN 0002-7863 (<https://search.worldcat.org/issn/0002-7863>).
6. "Synvista Therapeutics Inc. - BioCentury Company Profiles - BCIQ" (https://profiles.biocentury.com/companies/synvista_therapeutics_inc). *BioCentury*. Retrieved 2023-05-30.
7. "Alteon 10-K For the fiscal year ended December 31, 1996" (<https://www.sec.gov/Archives/edgar/data/878903/0000893220-97-000614.txt>). Alteon via SEC Edgar. March 27, 1997.
8. Harry Keen; JH Fukker; G Menzinger (July 19, 1997). "Early closure of European Pimagedine trial" ([http://www.thelancet.com/journals/lancet/article/PIIS0140-6736\(97\)26029-0/fulltext](http://www.thelancet.com/journals/lancet/article/PIIS0140-6736(97)26029-0/fulltext)). *The Lancet*. **350** (9072). PlumX Metrics: 214–215. doi:10.1016/S0140-6736(97)26029-0 (<https://doi.org/10.1016%2FS0140-6736%2897%2926029-0>). PMID 9250200 (<https://pubmed.ncbi.nlm.nih.gov/9250200>). S2CID 54316555 (<https://api.semanticscholar.org/CorpusID:54316555>).
9. Barbara Marsh (January 3, 1998). "Biotech's New Watchword: Partnership" (<https://www.latimes.com/archives/la-xpm-1998-jan-03-fi-4444-story.html>). *Los Angeles Times*. Retrieved August 17, 2017.
10. "Alteon May Drop Pimagedine In NIDDM" (<http://www.thepharmaletter.com/file/19466/alteon-may-drop-pimagedine-in-niddm.html>). The Pharma Letter. March 19, 1998. Retrieved August 17, 2017.
11. "Alteon Shares Plummet On Poor Pimagedine Test Results" (<http://www.sddt.com/News/article.cfm?SourceCode=19981116faq>). San Diego Source. November 16, 1998. Retrieved August 17, 2017.
12. "Genentech pulls out of pimagedine deal" (<https://www.thepharmaletter.com/genentech-pulls-out-of-pimagedine-deal>). *www.thepharmaletter.com*. 1999-02-10.
13. "Alteon's pimagedine fails primary endpoint" (<http://www.thepharmaletter.com/file/71571/alteons-pimagedine-fails-primary-endpoint.html>). The Pharma Letter. April 12, 1999. Retrieved August 17, 2017.
14. "ALTEON, INC. FORM 10-K" (<https://www.sec.gov/Archives/edgar/data/878903/0000893220-00-000381.txt>) (TXT). 2000-03-31.
15. "FORM 10-K ALTEON INC" (<https://www.sec.gov/Archives/edgar/data/878903/000089322001000240/0000893220-01-000240.txt>) (TXT). 2001-03-07.
16. "ALTEON INC. FORM 10-K FOR FISCAL YEAR END 12/31/01" (<https://web.archive.org/web/20130313201923/http://www.sec.gov/Archives/edgar/data/878903/000089322002000222/0000893220-02-000222.txt>). Archived from the original (<https://www.sec.gov/Archives/edgar/data/878903/000089322002000222/0000893220-02-000222.txt>) on 2013-03-13.
17. "Form 10-K for Alteon Inc" (<https://web.archive.org/web/20130313201919/http://www.sec.gov/Archives/edgar/data/878903/000089322003000272/0000893220-03-000272.txt>). Archived from the original (<https://www.sec.gov/Archives/edgar/data/878903/000089322003000272/0000893220-03-000272.txt>) on 2013-03-13.